Scalability & Future Plan

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Project Name	EcoTrack - AI-Powered Carbon Footprint Tracker
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the EcoTrack solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application is not yet deployed to a public hosting platform	Limits accessibility to local testing only; real-world users cannot access EcoTrack yet.	High
2	Single-instance Flask backend without load balancing	May struggle to handle high concurrent traffic if user base grows significantly.	Medium
3	AI suggestion generation depends on the external Groq API	Response times and availability are affected during peak load or third-party API downtime.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Flask development server, tested locally with up to 100 concurrent virtual users.	Deploy on a cloud platform (Render/AWS) using a production WSGI server (Gunicorn) behind a load balancer with horizontal scaling.
2	Data Storage	Firebase Firestore (NoSQL) with a local SQLite fallback for development.	Optimize Firestore indexing and introduce data archiving/partitioning strategies as historical habit logs grow.
3	Performance	Synchronous calls to the Carbon Interface and Groq APIs for every request.	Introduce caching (e.g., Redis) for repeated emission-factor lookups and asynchronous/background processing for AI suggestions.
4	Security	Firebase Authentication for user sign-up and login.	Add rate limiting, API key rotation, and role-based access control for future admin/analytics features.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Public deployment (Render/Vercel with production database)	1 month post-submission	Makes EcoTrack accessible to real users beyond the local demo.
Phase 2	Social sharing of badges and leaderboards	2 months	Increases user engagement and strengthens the gamification loop.
Phase 3	Mobile app version (React Native)	3-4 months	Expands reach to mobile-first users for on-the-go daily habit logging.
Phase 4	ML-based personalized carbon reduction targets	6 months	Improves suggestion accuracy and long-term user retention through smarter personalization.